Worksheet 4.6 — Discrete Random Variables and the Binomial Distribution

Mathematics: Analysis & Approaches SL (IB) · Unit 4: Statistics and Probability

The Big Ideas

The binomial distribution models a fixed number of independent yes/no trials.

- $E(X) = \sum xP(X = x)$.
- $X \sim B(n, p)$: $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$, $E(X) = np$, $\mathrm{Var}(X) = np(1-p)$.

Worked Examples

Example 1 — Unknown table entry

$P(X = 0) = 0.1$, $P(X = 1) = k$, $P(X = 2) = 0.5$. Find k and $E(X)$.

Solution. $k = 1 - 0.1 - 0.5 = 0.4$. $E(X) = 0(0.1) + 1(0.4) + 2(0.5)$.

Conclusion: $k = 0.4$, $E(X) = 1.4$.

Example 2 — Binomial probability

$X \sim B(9, 0.35)$. Find $P(X = 3)$.

Solution. $\binom{9}{3}(0.35)^3(0.65)^6 \approx 84(0.042875)(0.0754) \approx 0.272$.

Conclusion: $P(X = 3) \approx 0.272$.

Example 3 — At least one

$X \sim B(6, 0.15)$. Find $P(X \geq 1)$.

Solution. $P(X = 0) = (0.85)^6 \approx 0.377$.

Conclusion: $P(X \geq 1) \approx 0.623$.

Example 4 — Solve for p

$X \sim B(5, p)$ and $P(X = 5) = 0.00243$. Find p.

Solution. $p^5 = 0.00243 \Rightarrow p = 0.00243^{1/5} = 0.3$ (recognizing $0.3^5 = 0.00243$).

Conclusion: $p = 0.3$.

Example 5 — Expected value of a custom discrete variable

A game pays \$10 with probability 0.2, \$0 with probability 0.5, and −\$5 (a loss) with probability 0.3. Find the expected value.

Solution. $E(X) = 10(0.2) + 0(0.5) + (-5)(0.3) = 2 + 0 - 1.5$.

Conclusion: $E(X) = \$0.50$ — the game is favourable on average.

Example 6 — At most k successes

$X \sim B(6, 0.25)$. Find $P(X \leq 1)$.

Solution. $P(X = 0) = (0.75)^6 \approx 0.178$. $P(X = 1) = \binom{6}{1}(0.25)(0.75)^5 \approx 0.356$.

Conclusion: $P(X \leq 1) \approx 0.178 + 0.356 = 0.534$.

Example 7 — Between two values

$X \sim B(10, 0.4)$. Find $P(2 \leq X \leq 4)$.

Solution. $P(2) = \binom{10}{2}(0.4)^2(0.6)^8 \approx 0.121$; $P(3) \approx 0.215$; $P(4) \approx 0.251$.

Conclusion: $P(2 \leq X \leq 4) \approx 0.121 + 0.215 + 0.251 = 0.587$.

Example 8 — Solve for n given a mean

$X \sim B(n, 0.3)$ and $E(X) = 12$. Find n.

Solution. $np = 12 \Rightarrow 0.3n = 12$.

Conclusion: $n = 40$.

Example 9 — Application — warranty claims

A store sells 20 items with a 5% individual chance of a warranty claim. Find the expected number of claims and the probability of at least 2 claims.

Solution. $E(X) = 20(0.05) = 1$. $P(X \geq 2) = 1 - P(0) - P(1) = 1 - (0.95)^{20} - 20(0.05)(0.95)^{19} \approx 1 - 0.358 - 0.377$.

Conclusion: expected claims $= 1$; $P(X \geq 2) \approx 0.265$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

$P(X = 0) = 0.2$, $P(X = 1) = k$, $P(X = 2) = 0.35$. Find k and $E(X)$.

Question 2

$X \sim B(7, 0.4)$. Find $P(X = 2)$.

Question 3

$X \sim B(8, 0.1)$. Find $P(X \geq 1)$.

Question 4

$X \sim B(40, 0.25)$. Find the mean and variance.

Question 5 — Challenge

A basketball player makes 65% of free throws; for 8 attempts, find P(makes at least 6).

Question 6

$X \sim B(4, p)$ and $P(X = 0) = 0.0016$. Find p.

Question 7 — Challenge

A factory tests 10 items with a 4% individual defect rate; find $P(\text{at most 1 defective})$.

Question 8

A game pays \$20 with probability 0.15, \$0 with probability 0.6, and $-$\$8 with probability 0.25. Find the expected value.

Question 9

$X \sim B(7, 0.3)$. Find $P(X \leq 1)$.

Question 10

$X \sim B(12, 0.35)$. Find $P(3 \leq X \leq 5)$.

Question 11

$X \sim B(n, 0.2)$ and $E(X) = 15$. Find n.

Question 12 — Challenge

A shop sells 25 items with a 4% individual chance of a return; find the expected number of returns and $P(\text{at least 2 returns})$.

Question 13 — Challenge

$X \sim B(6, p)$ and $\text{Var}(X) = 1.5$ with $p < 0.5$; find p (solve $6p(1-p) = 1.5$).

Question Recap & Answer Key

Question Recap

1. $P(X=0)=0.2$, $P(X=1)=k$, $P(X=2)=0.35$. Find k and $E(X)$.
2. $X \sim B(7, 0.4)$. Find $P(X=2)$.
3. $X \sim B(8, 0.1)$. Find $P(X \geq 1)$.
4. $X \sim B(40, 0.25)$. Find the mean and variance.
5. A basketball player makes 65% of free throws; for 8 attempts, find P(makes at least 6).
6. $X \sim B(4, p)$ and $P(X=0)=0.0016$. Find p.
7. A factory tests 10 items with a 4% individual defect rate; find P(at most 1 defective).
8. A game pays \$20 with probability 0.15, \$0 with probability 0.6, and −\$8 with probability 0.25. Find the expected value.
9. $X \sim B(7, 0.3)$. Find $P(X \leq 1)$.
10. $X \sim B(12, 0.35)$. Find $P(3 \leq X \leq 5)$.
11. $X \sim B(n, 0.2)$ and $E(X)=15$. Find n.
12. A shop sells 25 items with a 4% individual chance of a return; find the expected number of returns and P(at least 2 returns).
13. $X \sim B(6, p)$ and $\text{Var}(X)=1.5$ with $p<0.5$; find p (solve $6p(1-p)=1.5$).

✓ Answer Key

1. $k=0.45$; $E(X)=0.45+0.7=1.15$
2. $\binom{7}{2}(0.4)^2(0.6)^5 \approx 21(0.16)(0.07776) \approx 0.261$
3. $P(X=0)=(0.9)^8 \approx 0.4305$; $P(X \geq 1) \approx 0.5695$
4. mean $=10$, variance $=40(0.25)(0.75)=7.5$
5. $P(6)+P(7)+P(8)=\binom{8}{6}(0.65)^6(0.35)^2+\binom{8}{7}(0.65)^7(0.35)+\binom{8}{8}(0.65)^8 \approx 0.2587+0.1373+0.0319 \approx 0.428$
6. $(1-p)^4=0.0016 \Rightarrow 1-p=0.0016^{1/4}=0.2 \Rightarrow p=0.8$
7. $P(0)=(0.96)^{10} \approx 0.6648$; $P(1)=\binom{10}{1}(0.04)(0.96)^9 \approx 0.2770$; sum ≈ 0.9418
8. $E(X)=20(0.15)+0(0.6)+(-8)(0.25)=3-2=\$1.00$
9. $P(0)=(0.7)^7 \approx 0.0824$; $P(1)=\binom{7}{1}(0.3)(0.7)^6 \approx 0.2471$; sum ≈ 0.3295
10. $P(3) \approx 0.2039$, $P(4) \approx 0.2367$, $P(5) \approx 0.1959$; sum ≈ 0.6365
11. $0.2n=15 \Rightarrow n=75$
12. $E(X)=25(0.04)=1$; $P(X \geq 2)=1-(0.96)^{25}-25(0.04)(0.96)^{24} \approx 1-0.360-0.375 \approx 0.265$
13. $6p-6p^2=1.5 \Rightarrow 6p^2-6p+1.5=0 \Rightarrow 4p^2-4p+1=0 \Rightarrow (2p-1)^2=0 \Rightarrow p=0.5$ (boundary; for $p<0.5$ strictly, note this is the only solution — the variance is maximized exactly at $p=0.5$)